STEADY STATE STABILITY

Module 1

STEADY STATE STABILITY

- It is important to know the conditions of stable operation of an electric drive when its equilibrium state is disturbed.
- Two types of disturbances are normally considered.

- one where the **change from the equilibrium state take place slowly** and the effect of either the inertia of rotating masses or that of inductance is not significant – the study of stability under conditions in this case relate to the field of **steady state stability**.

- sudden and fast change from the equilibrium state, as a result of which the effect of neither the inertia nor the inductance can be ignored - the study of stability under conditions in this case relate to the field of **dynamic or transient stability**.

- **Equilibrium speed** of a motor-load system is obtained when motor torque is equal to load torque.
- The drive will operate in steady state at this speed, provided it is the speed of stable equilibrium.
- **Stability** of motor-load combination is defined as the capacity of the system which enables it to develop forces which restore equilibrium.
- All drives does not possess steady state stability.

STEADY STATE STABILITY

- The state of equilibrium may be stable, unstable or neutral.
- The drive may come out of the equilibrium state under the influence of any unexpected external factors like change in the input to the motor or due to internal factors like change in load.

  - in the case of stable equilibrium the drive returns to its steady state.

  - in the case of unstable equilibrium the speed of the drive either increases tremendously or decreases and comes to rest.

  - in the case of neutral state, the drive preserves its steady state at different speeds within a small range.

- The figure shows 4 combinations of motor torque and load torque

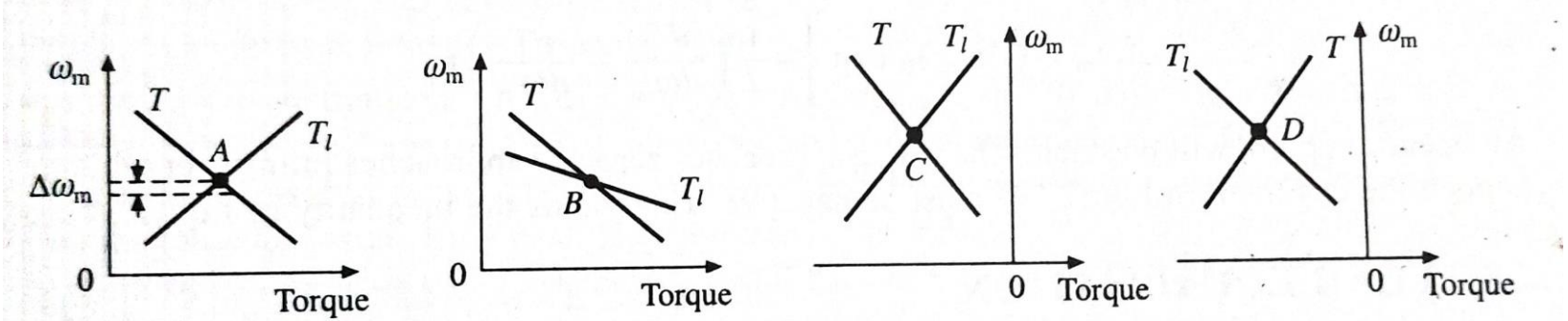
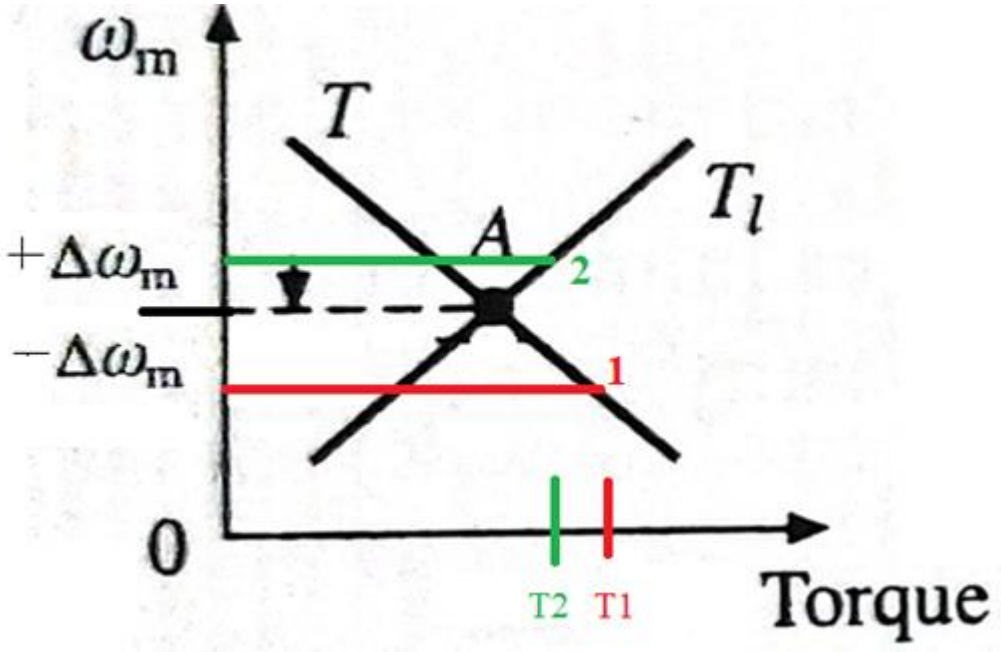


Fig. 2.9 Points A and C are stable and B and D are unstable

STEADY STATE STABILITY(Contd…..)

- Consider the speed-torque characteristics given below. Let us consider the steady state stability of equilibrium point A.
- The equilibrium point is termed as stable when the operation will be restored to it after a small departure from it due to a disturbance in the motor or load.
- Let the disturbance cause a reduction of $\Delta\omega_m$

in speed. At new speed corresponding to point 1, the motor torque(T1) is > load torque. Now the motor will accelerate and operation will be restored to point A.



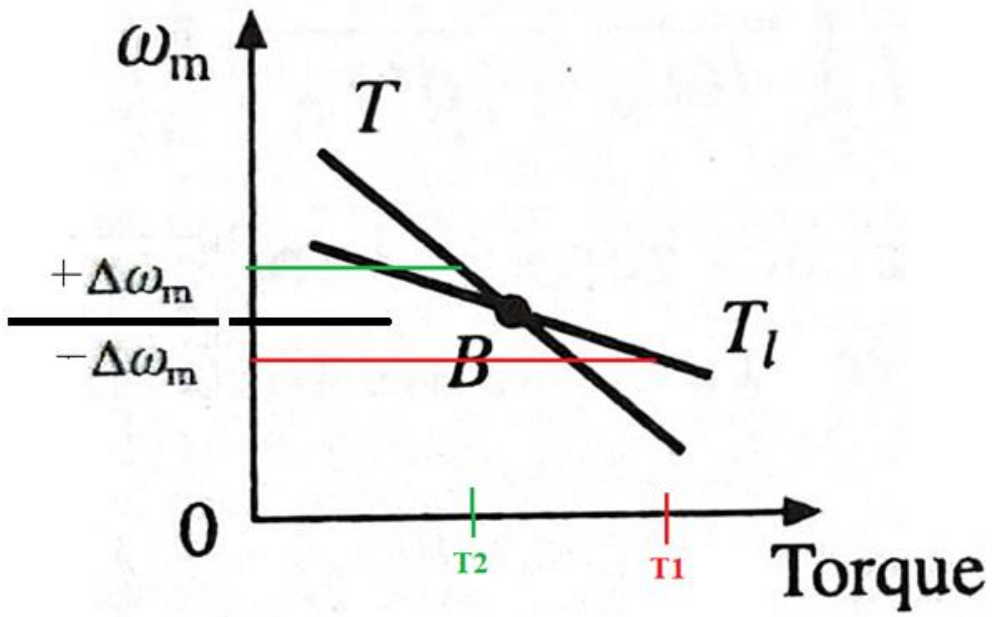
- Let the disturbance cause an increase of $\Delta\omega_m$

in speed. At new speed corresponding to point 2, the load torque(T2) is > motor torque. Now the motor will decelerate and operation will be restored to point A.

- Hence the drive is steady-state stable at operating point A

STEADY STATE STABILITY (Contd….)

- Now consider the equilibrium point B which is obtained when the same drive operates another load.
- A decrease in speed causes the load torque to be greater than the motor torque. Now the drive decelerates and the operating point moves away from B.
- An increase in speed causes the motor torque to be greater than the load torque. Now the drive accelerates and the operating point moves away from B.
- Thus B is an unstable point of equilibrium.



STEADY STATE STABILITY (Contd….)

Criteria for steady state stability:-

- An accurate study of the stability of the drives requires the solution of differential equation of the drive system.
- If the transient portion of the solution of these equations approaches zero with increasing time, then the effect of any disturbance will die away and the system is stable.
- In the absence of any such analysis, the stability problem is analysed by a study of the steady state speed-torque curves of both motor & load, by assuming that all departures from equilibrium will be along these curves.
- An equilibrium point will be stable when an increase in speed causes load torque to exceed motor torque.
- i.e., at equilibrium $\frac{dT_l}{d\omega_m} > \frac{dT}{d\omega_m}$ ……..(1) is satisfied.

STEADY STATE STABILITY (Contd….)

- Let T = equilibrium value of motor torque

  T_l = equilibrium value of load torque

  ω_m= equilibrium value of motor speed

  ΔT = small change in motor torque

  ΔT_l = small change in load torque

  $\Delta\omega_m$ = small change in motor speed

After a small change in equilibrium the torque equation becomes

$$(T + \Delta T) = (T_l + \Delta T_l) + J\frac{d(\omega_m + \Delta\omega_m)}{dt} \quad \ldots\ldots\ldots\ldots\ldots(2)$$

or

$$= T_l + \Delta T_l + J\frac{d\omega_m}{dt} + J\frac{d\Delta\omega_m}{dt} \quad \ldots\ldots\ldots\ldots\ldots(3)$$

- But $T = T_l + J\frac{d\omega_m}{dt}$ ……(4) and eqn (3) can be written as

$$T + \Delta T = T + \Delta T_l + J\frac{d\Delta\omega_m}{dt}$$

or

$$J\frac{d\Delta\omega_m}{dt} = \Delta T - \Delta T_l \quad \ldots\ldots\ldots\ldots\ldots(5)$$

STEADY STATE STABILITY (Contd….)

$$J\frac{d\Delta\omega_m}{dt} + \Delta T_l - \Delta T = 0 \quad \text{……(6)}$$

- Equation (6) is the torque in which all quantities are expressed in terms of the deviation from equilibrium values.
- If it is assumed that these increments are so small that they may be expressed as linear function of the change in speed. Then

$$\Delta T = \left(\frac{dT}{d\omega_m}\right)\Delta\omega_m \quad \text{……..(7)}$$

and $\Delta T_l = \left(\frac{dT_l}{d\omega_m}\right)\Delta\omega_m$ ……(8)

where $\left(\frac{dT}{d\omega_m}\right)$ and $\left(\frac{dT_l}{d\omega_m}\right)$ are the slopes of the steady-state speed-torque curves of the motor and the load respectively at the operating point under consideration.

- Substituting eqn (7) & (8) in (6) and rearranging

$$J\frac{d\Delta\omega_m}{dt} = \Delta T - \Delta T_l$$

$$= \left(\frac{dT}{d\omega_m}\right)\Delta\omega_m - \left(\frac{dT_l}{d\omega_m}\right)\Delta\omega_m = \left[\left(\frac{dT}{d\omega_m}\right) - \left(\frac{dT_l}{d\omega_m}\right)\right]\Delta\omega_m$$

or $J\frac{d\Delta\omega_m}{dt} + \left[\left(\frac{dT_l}{d\omega_m}\right) - \left(\frac{dT}{d\omega_m}\right)\right]\Delta\omega_m = 0$ ……..(9)

STEADY STATE STABILITY (Contd….)

- Eqn (9) is a first order linear differential equation.
- If initial deviation in speed at t=0 be $\Delta\omega_{m0}$ then the solution of eqn (9) will be

$$\Delta\omega_m=(\Delta\omega_m)_0 \exp\left\{-\frac{1}{J}\left[\left(\frac{dT_l}{d\omega_m}\right)-\left(\frac{dT}{d\omega_m}\right)\right] t \right. \quad \text{.....(10)}$$

- For the system to be stable, the exponent of eqn (10) must be –ve. Then the speed increment will disappear with time and the system will return to its equilibrium speed.
- If the exponent is greater than zero, the speed deviation will increase with time and the system will move away from equilibrium.
- If the exponent is equal to zero, the equation is insufficient to discuss stability.
- The exponent is –ve if $\frac{dT_l}{d\omega_m}-\frac{dT}{d\omega_m}>0$.
- Thus for a decrease in speed motor torque must exceed load torque and for an increase in speed the motor torque must be less than load torque.

STEADY STATE STABILITY (Contd….)

- The stability of operating point is evaluated by assuming a small change in speed away from the equilibrium point.
- With this speed increment assumed, the relative values of motor & load torques will determine whether the speed will return to its previous value.
- i.e., if an increase in speed brings about a greater increase in load torque than motor torque, the speed will tend to decrease and return to its original value which is then a stable point.
- Conversely when the speed continues to increase the system is unstable.
- The criterion for stability is an approximation in the sense that it is assumed that the steady state speed-torque curves are traversed when small disturbances cause a departure in speed from some operating point.
- This assumption may or may not be valid, as in many cases the transient speed-torque relations are different from the steady state curve.